



COURSE HANDOUT (PART-II)

Date: 01-08-2026

In addition to part I (General Handout for all courses appended to the time table) this portion gives further specific details regarding the course.

Course No. : CHEM F313

Course Title : Instrumental Methods of Analysis

Instructor-in-charge : Ajay Kumar Sah

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Practical Instructors : Ajay Kumar Sah, Mrinmoyee Basu, Shubhangi Sharma, Vineeta Chaturvedi, Ram Kushal Verma, Shruti Soni, Priyanka Kumari, Ghanshyam Sharma, Prakriti Sharma, Aditya Mohan Shukla, Prekshi Gupta, Samya Suvra Datta, Sneha Mishra, Aparna Sharma, Ayushi Tiwari, Shiyanshi Nuwal

Course Description: Principles and practice of modern instrumental methods of chemical analysis. Emphasis on spectroscopic techniques such as UV Visible, infrared, NMR (^1H , ^{13}C and other elements, NOE, correlation spectroscopies), ESR, atomic absorption and emission, photoelectron, Mössbauer, and fluorescence. Other topics will include mass spectrometry, separation techniques, light scattering, electroanalytical methods, thermal analysis, and diffraction methods.

Scope and Objective of the Course: Chemists extensively use modern sophisticated electronic instruments in various areas such as chemical analysis, structure determination, identification of reaction pathways and rates etc. This course aims to introduce the basic theory and experimental details of such chemical instrumentation. Some of the popular absorption spectroscopic techniques such as UV-visible, IR, NMR, mass spectrometry etc. will be discussed in detail.

Text Books: T1. Kemp W., 'Organic Spectroscopy', 3rd ed., Palgrave, New York (1991).

Reference Books: R1. Willard H. H., Merritt Jr. L. L., Dean J. A., Settle F. A. S., "Instrumental Methods of Analysis", 7th Ed., Wadsworth, 2009, Cengage Learning India Pvt. Ltd. Fifth Indian reprint by CBS Publishers & Distributors Pvt. Ltd.

R2. Silverstein R. M., and Webster F. X., "Spectrometric Identification of Organic Compounds", 6th ed., John Wiley & Sons, New York (1998).

R3. Skoog D. A., Holler F. J., and Crouch S. R., "Principles of Instrumental Analysis", 6th ed., Thomson Brooks/Cole, Cengage Learning, New Delhi (2007).

Course Plan: A. Lecture Sessions:



Module No.	Lecture Session	Reference: Chap./Sec. (Book)	Learning Outcome
1 (Lecture 1)	Introduction to Chemical analysis, Instrumental methods Measurements, Signals and Data	Ch 1-2 (R1)	Introduction to the field of analysis and electromagnetic radiation
2 (Lecture 2)	Introduction to Energy and Electromagnetic spectrum, Absorption and emission spectroscopy.	Ch.1 (T1) Ch 5 (R1)	
3 (Lecture 3-7)	Ultraviolet (UV) and visible spectroscopy: Light Absorption, theory, instrumentation	Ch. 4.1-4.3(T1)	Ability to choose systems to apply the UV-Vis spectroscopy and concept of analysis
	UV-Visible Spectroscopy: Solvents, applications	Ch. 4.4-4.10(T1)	
	UV-visible Spectroscopy: stereo chemical factors, quantitative electronic spectroscopy	Ch. 4.11-4S.1(T1) Ch 6-7 (R1)	
	Fluorescence and Phosphorescence	Ch. 4S.2 (T1) Ch. 8 (R1)	Concept on emission spectroscopy.
	UV-Visible Spectroscopy: CT complexes, symmetry, ORD-CD	Ch. 4S.3-4S.6(T1)	Understanding about the optical property with related to optical activity
4 (Lecture 8-11)	Infrared Spectroscopy: Molecular vibrations; related factors	Ch. 2.1-2.3(T1)	Understanding of the working principle of vibrational spectroscopy and analysis of the spectral data
	Infrared spectroscopy: Instrumentation, and Applications	Ch. 2.4-2.8 (T1) Ch 11 (R1)	
	Raman spectroscopy	Ch.2S.3(T1) Ch 12 (R1)	Understanding of the basic principle of Raman spectroscopy
5 (Lecture 12-16)	Mass spectrometry: Basics, Instrumentation, Isotopic abundance, and Molecular ion. meta-stable ions, fragmentation processes	Ch 5 (T1)	Understanding of the working principle mass spectroscopy and analysis of organic compounds based on fragmentation
6 (Lecture 17-20)	X-ray Methods: X-ray absorption, X- ray fluorescence, X-ray Diffractions	Ch 13 (R1) Ch. 8-9(R3) Lecture notes	Introduction to principle and application of X-ray absorption



7 (Lecture 21-22)	Flame Emission & Atomic Absorption Spectroscopy	Ch 9 (R1)	Quantitative analysis for different metals <i>via</i> spectroscopy
8 (Lecture 23-24)	Thermal Analysis: Differential scanning calorimetry, Thermogravimetry	Ch. 25(R1) Ch. 31 (R3)	Concept building on the thermal analysis to understand physical and chemical processes of compounds
9 (Lecture 25-26)	Radiochemical/ Nuclear Methods Neutron activation analysis	Ch 14 (R1)	Knowledge about the process of analysis of radioactive materials
10 (Lecture 27-29)	Separation Techniques: Chromatography – Basics and applications, HPLC, GC etc.	Ch 17-20 (R1) Ch 27-30 (R3)	Understanding about the separation techniques involving basics of chemistry
11 (Lecture 30-31)	Hyphenated techniques, Applications such as GC-MS, MS-MS, ICP-MS, isotopic substitution etc.	Ch. 5S.1-5S.5(T1)	Knowledge about few advanced techniques
12 (Lecture 32-33)	Electronic Paramagnetic Resonance (EPR)	Class notes	Analysis of organic radicals
13 (Lecture 34-40)	Nuclear Magnetic Resonance (NMR) spectroscopy: Understanding Magnetic Resonance phenomena and the concept of ^1H NMR, ^{13}C NMR spectroscopy, NOE and correlation spectroscopies.	Ch 3 (T1) Ch 15 (R1) Class notes	Concept building on the theory of magnetic resonance and the applications of NMR in the structure elucidation of organic compounds

B. Practical Sessions:

Regular sessions: (8) In these sessions the student (in groups) will perform an experiment on the techniques such as UV-vis spectroscopy, Spectrofluorimetry, IR spectroscopy, Flame photometry, Polarimetry, DSC-TGA and HPLC analysis. Among some more instrumental techniques like high voltage electrophoresis, NMR, GC analysis, HRMS, few could be picked up for demonstration purpose or could be in list of experiments to be performed. The instructors will make available experimental handout sheets for each of these laboratory experiments.



All students are required to

1. Write a report of the experiment
2. Come prepared for each of the experiments respectively.

USE OF APPROPRIATE PERSONAL PROTECTIVE EQUIPMENT (PPE) (e.g. Apron, gloves, covered footwear etc.) IS MANDATORY DURING ALL LABORATORY SESSIONS. DEFAULTERS WILL NOT BE ALLOWED TO PERFORM WITHOUT PPE.

Evaluation Scheme

Total marks: 300

A. Theory (210 Marks)

Component	Duration	Marks	Date & Time	Nature of component (Close Book/ Open Book)
Quizzes*	20 min each	40	Continuous	Close Book
Attendance based evaluation**	10 min each	20	Continuous	Close Book
Mid-Semester Test	90 min.	50	10/10 FN1	Close Book
Comprehensive Examination [§]	180 min.	100	15/12 FN	Close Book and Open Book

*THREE quizzes (20 Marks each) will be conducted in entire semester. Best TWO will be considered for final evaluation.

**FOUR tutorial tests will be conducted in the entire semester. The tutorial tests will be graded only if the student maintains a minimum of 50% attendance in lectures and tutorials.

[§]The comprehensive examination will have a closed book portion for 25 marks, and an open book section for 75 marks.

Minimum criterion for awarding valid grades: Where a student continued to remain registered in a course but gave the instructor inadequate opportunity to evaluate him/her by absenting himself/herself from quizzes/ tests/ examinations/other components of evaluation, or by appearing in the same for the sake of appearance without applying himself/herself to the task in hand or by submitting a blank script (answer book) this event will be reported as NC (Not Cleared). **Moreover, a student should obtain 30% of the average of the top 10 performers in the class, or 40% of the median marks of the class, whichever is lower to clear the course.** If any student gets the marks lower than the prescribed standard mentioned above, he/she may be awarded NC.



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B. Practical (90 Marks): Ten marks each for the regular sessions followed by a test of 10 marks at the end of the regular lab class.

Chamber Consultation Hours: Saturday 10-11 am

Makeup Policy: See Part I for details. However, it may be noted that **it is impossible to arrange make up of practical sessions and for Tutorial Evaluations.**

Notices: It will be displayed on the **Nalanda and/or Department of Chemistry Notice board**

Ajay Kumar Sah

Instructor-in-charge
CHEM F313